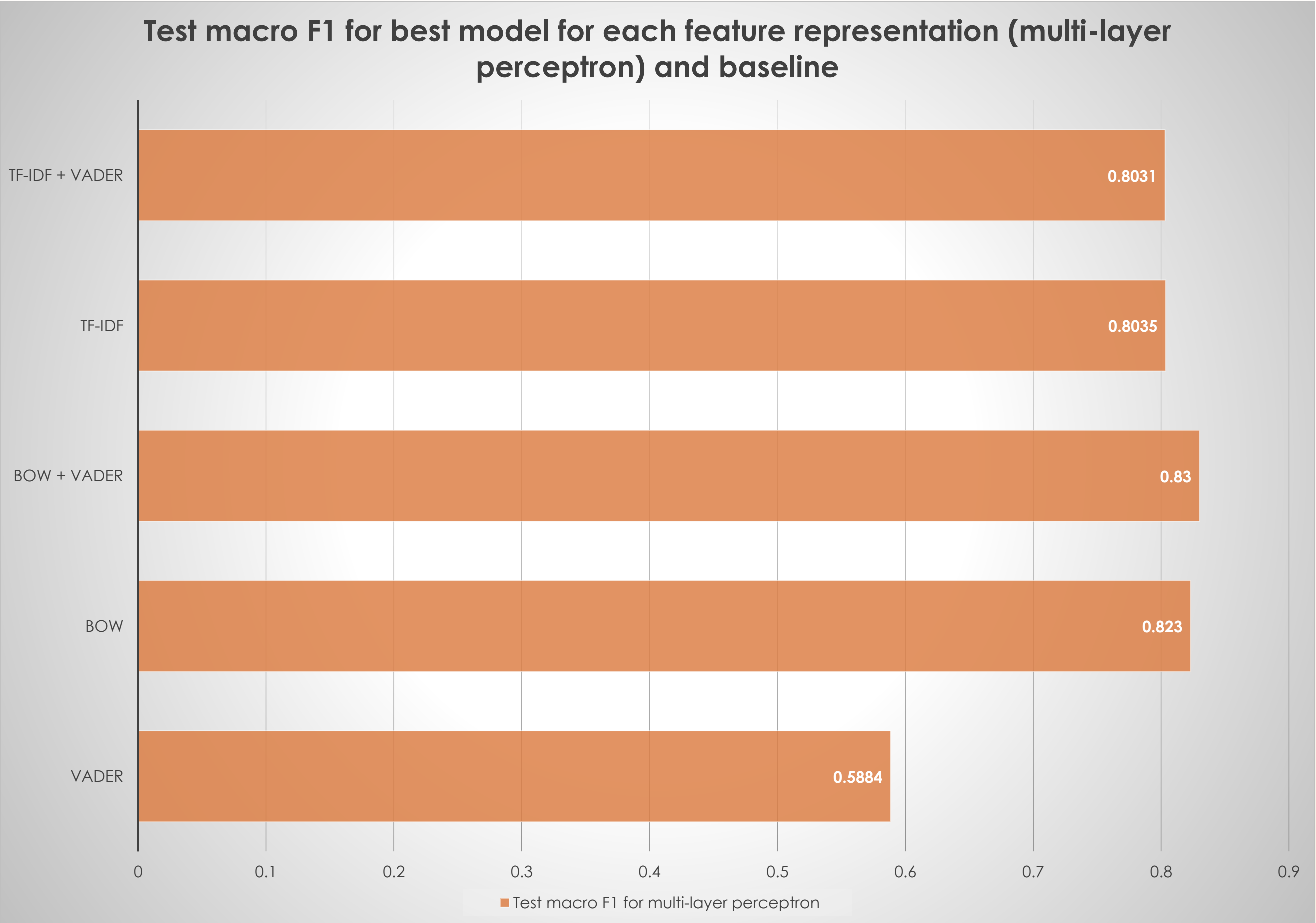
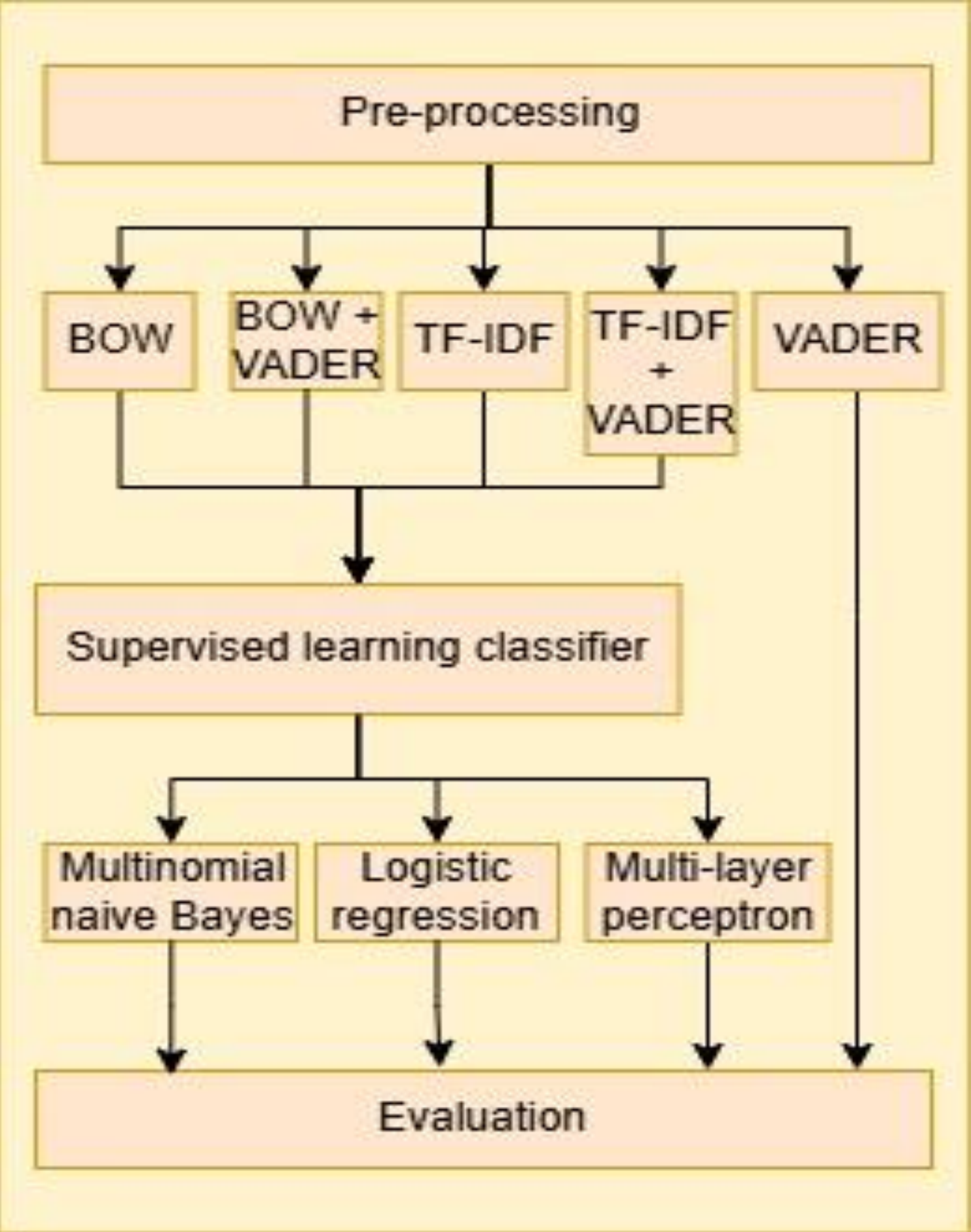


# Sentiment classification of patient medication feedback text

Does combining VADER lexicon with conventional feature representations (BOW, TF-IDF) improve the performance of sentiment classification by adding lexicon information?



**Important information:**

- 10-scale rating is converted to 2-scale sentiment label (1-4: negative, 5-10: positive)
- Preprocessing steps contain regular expression cleaning, lowercasing, tokenization, part of speech tagging, lemmatisation and stop word removal

| Feature set                            |          |       | Algorithm               | Test macro average F1 |
|----------------------------------------|----------|-------|-------------------------|-----------------------|
| VADER (baseline)                       | compound | score | -                       | 0.5884                |
| BOW                                    |          |       | Multinomial naïve Bayes | 0.7690                |
|                                        |          |       | Logistic regression     | 0.7950                |
|                                        |          |       | Random forest           | 0.4075                |
|                                        |          |       | Multi-layer perceptron  | 0.8230                |
| BOW + VADER pos, neu and neg scores    |          |       | Multinomial naïve Bayes | 0.7695                |
|                                        |          |       | Logistic regression     | 0.7920                |
|                                        |          |       | Multi-layer perceptron  | 0.8300*               |
| TF-IDF                                 |          |       | Multinomial naïve Bayes | 0.7296                |
|                                        |          |       | Logistic regression     | 0.7970                |
|                                        |          |       | Multi-layer perceptron  | 0.8035                |
| TF-IDF + VADER pos, neu and neg scores |          |       | Multinomial naïve Bayes | 0.7342                |
|                                        |          |       | Logistic regression     | 0.7968                |
|                                        |          |       | Multi-layer perceptron  | 0.8031                |

**Key Findings**

- For all feature representations, all three models outperformed the baseline VADER model by at least 0.1412 in terms of macro average F1
- For all feature representations, multi-layer perceptron outperformed other machine learning models with lead of 0.0063-0.0380 in terms of macro average F1
- The best performing model is multi-layer perceptron with BOW +VADER feature representation, with test macro average F1 of 0.8300
- For logistic regression, TD-IDF outperformed BOW because low magnitude of TF-IDF favours logistic regression
- For multinomial naïve Bayes and multi-layer perceptron, BOW outperformed TF-IDF because multinomial naïve Bayes is designed for raw count and multi-layer perceptron makes use of the word frequency to extract hidden pattern
- Adding VADER to BOW and TF-IDF may improve or degrade the performance (50% of the cases improved whereas 50% of the cases degraded)
- Suspected reason for degradation is VADER introduced noise to the model